

be used to burn a hex file through USB using ATMEL Flip software.

Press S3 (HWB) button and S2 reset button simultaneously and then first release S2 while S3 is pressed. This will put the board into an inbuilt bootloader programming mode. In programming mode, the board can be seen in device manager of the PC. You

can load the hex code into the MCU using ATMEL Flip programmer.

But to make a generic code run with read and write support in Flash memory, you need to erase the entire Flash (default bootloader) and load a new bootloader called 'ATMEL USB DFU Bootloader' from ATMEL's website. This also enables the micro-

controller to run at its full speed of 16 MHz. You just need to make necessary changes in the fuses to make it run with external clock of 16 MHz (without CKDIV8 fuse enabled). Now burn the new bootloader using any ISP programmer by connecting the programmer to the ISP connector on the board.

*User-interface program.* After as-

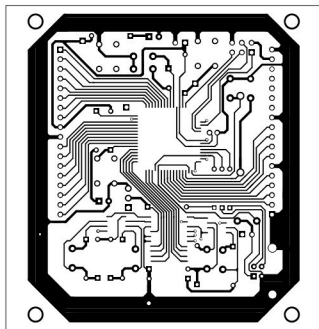


Fig. 3: An actual-size, double-side, solder-side PCB track layout of USB data acquisition system

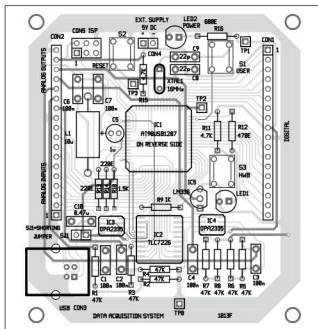


Fig. 5: Component layout

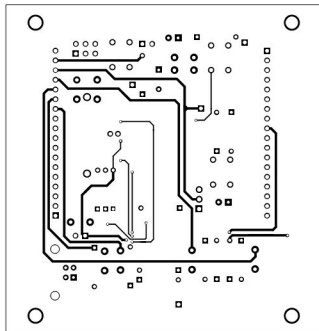


Fig. 4: Component-side track layout

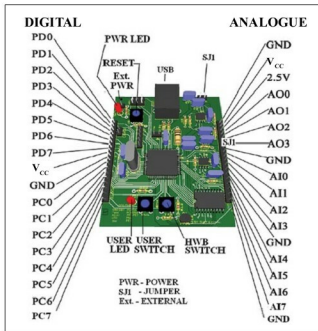


Fig. 6: The author's assembled board with input and output pin details